ENEL 573 Computer Networks: Sample Midterm Exam (Fall 2026)

Name: _____ UCID: _____ Score: _____

Problem 1

Suppose users share a 2-Mbps link. Also suppose each user requires 300 Kbps when transmitting, but each user only transmits 12 percent of the time.

a. When circuit switching is used, how many users can be supported?

b. For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.

c. Suppose there are *M* users. Find the probability that at any given time, *n* users are transmitting simultaneously.

d. Find the probability that there are 7 or more users transmitting simultaneously.

Solution:

a. When circuit switching is used, only $\left\lfloor \frac{2Mbps}{300kbps} \right\rfloor = \lfloor 6.67 \rfloor = 6$ users can be supported.

b. Since a user is only active 12% of the time, hence the probability p that a user is transmitting, is 0.12

c. Given that the total number of users is M, assuming that the users are all independent and given p (q=1-p). The probability of having n of them transmitting at the same time is: $C_n^M \cdot p^n \cdot q^{M-n}$

d. To find the probability that 7 or more users are transmitting, first find the probability that 6 or less users are transmitting:

Let : P_A=P(6 or less users are transmitting)= $\sum_{k=0}^{6} C_k^M p^k q^{M-k}$

P(7 or more users are transmitting) = 1- P_A

Alternative Solution: P(7 or more users are transmitting) = $\sum_{k=7}^{M} C_k^M p^k q^{M-k}$

Problem 2

In an M/M/1 queue, customers arrive at the rate of λ = 15 per hour. What is the minimum server rate to ensure that

1) the sever is idle at least 10% of the time

2) the expected value of the queue length is not to exceed 10?

3) the probability that at least 20 people in the queue is at most 50%.

Solution:

1. the sever is idle at least 10% of the time:

For $P_0 \geq 0.10 \Rightarrow 1-\rho \geq 0.10 \Rightarrow \frac{\lambda}{\mu} \leq 0.9 \Rightarrow \mu \geq 16.67$

2. the expected value of the queue length is not to exceed 10:

$$E(N_q) = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)} \leq 10 \Rightarrow \mu = 16.375$$

3. the probability that at least 20 people in the queue is at most 50%:
$P_n = (1-\rho)\rho^n,\ n = 0, 1, 2, \cdots,$
$P(n \geq 21) = \sum_{n=21}^{\infty}(1-\rho)\rho^n \leq 0.5 \Rightarrow \mu \geq 15/e^{ln(0.5)/21}.$

Problem 3

1) Recall that with the CSMA/CD protocol, the adapter waits K× 512 bit times after a collision, where K is drawn randomly. For K = 100, how long does the adapter wait (in seconds) before starting to sense the medium for a 100 Mbps broadcast channel?

Basic idea: The wait time, Tw = K x 512 bit times.
For a 100 Mbps broadcast channel, 100 x 10^6 (= 10^8) bits are transmitted in 1 second.
Therefore, 1 bit time = 1/(10^8) seconds = 10^{-8} s.

Tw = 100 x 512 x 10^{-8} s. = 512 x 10^{-6} s.

Answer: The adapter waits for **512 x 10^{-6} s** before starting to sense the medium.

2) Consider the 5-bit generator, G=10011. If a receiver receives a data frame represented by the bit string 1010110 with CRC bits in it, was the frame received correctly? Justify your answer.

Basic idea: Divide the frame (= 1010110), including the CRC, by G (= 10011). If the remainder is 0000, then the frame has been received correctly. Otherwise, it has been incorrectly received.

```
Frame:    1010110
G:        10011
----------------------------------------------------------
          0011010   (The bit in the frame above the MSB of G was 1 so XOR frame and G.
           10011    (Shift G to right by 1 position.)
----------------------------------------------------------
          0011010   (The bit in the "frame" above the MSB of G was 0 so do NOT XOR frame and G.
            10011   (Shift G to right by 1 position.)
----------------------------------------------------------
          0001001   (The bit in the frame above the MSB of G was 1 so XOR frame and G.)
```

The computation stops now, because G has been shifted to the right-hand end of the "frame."
The remainder of the division is **1001**.

Answer: No, the frame was not received correctly by the receiver.